Experimental Studies on a 1-kW High-Gain *S*-Band Magnetron Amplifier With Output Phase Control Based on Load–Pull Characterization

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***Abstract*— A high power gain magnetron amplifier with output phase control based on load–pull characterization has been developed. The equivalent resonant cavity method was first introduced to facilitate the proposed magnetron amplifier system as a basic injection model, and the system behavior was qualitatively evaluated based on Alder's equation. It is the first time that a magnetron has been locked at a 58-dB gain. The magnetron amplifier shows an enhanced power-added efficiency of 72.0%, wide output phase control range, and 18.5-MHz bandwidth at a ratio of the output to the input power of 45 dB. This is the first time a successful magnetron output phase control based on load–pull at a power gain of 45 dB has been demonstrated. The magnetron output spectra were fully synchronized with a reference signal and the phase noise of the magnetron amplifier reached −106.3 dBc/Hz at an offset of 1 kHz. An output amplitude stability of 0.01 dB (peak to peak), a phase stability of 0.5° (peak to peak), and a high-efficiency dual 1-kW magnetron amplifier coherent-power combination based on the load–pull characterization was achieved. To the best of the authors' knowledge, this is one of the highest power gains and phase noise performances ever reported for an S-band injection-locked magnetron with a power output of over 1 kW.**

***Index Terms*— High-power amplifiers, injection-locked magnetron, load-pull, noise.**

I. Introduction

MICROWAVE processing materials have attracted a significant amount of attention in industry (in particular, in the field of chemical synthesis) because of their fast heating rate, high processing efficiency, and energy conservation properties. The heating pattern and uniformity of the materials during microwave heating is influenced by the microwave radiation frequency, so frequency selective heating or dynamic frequency and phase heating can be used to improve the heating uniformity or processing efficiency of materials under microwave irradiation [1]; furthermore, these techniques could be implemented using traveling wave tube amplifiers [2], klystrons [3], and related solid-state power amplifiers (SSPAs) [4]. For example, Kobayashi *et al.* [4] proposed an *S*-band 1-kW-class SSPA for wireless space applications with a drain efficiency of 57% and a power combining efficiency of 87%. Song *et al.* [2] realized a traveling wave tube with an output power of 1 kW and a dc-to-microwave conversion efficiency of over 40% in the *S*-band. These power amplifiers have an excellent performance for a variety of applications. However, they are expensive and thus not an ideal candidate for industrial heating applications.

Magnetrons, although regarded as noisy oscillators whose amplitude and phase fluctuate randomly, are widely used in industrial heating systems and household microwave ovens because of their low manufacturing costs and low energy consumption [5], [6]. Usually, a magnetron does not produce a monochromatic frequency, for example, microwave oven magnetrons are designed to be operated at 2450 MHz with a frequency tolerance limit of 50 MHz; they generate spurious noise in various frequency bands [7]. Therefore, others systems, such as Wi-Fi communication controls, Bluetooth equipment, and Zigbee sensors, suffer from electromagnetic interference in the vicinity of microwave heaters, especially in high-power microwave applications, on account of the leakage. However, a magnetron locked by a small injection signal, referred as a magnetron amplifier, is an effective way to solve the poor noise performance problem of magnetrons. In particular, this solution is very interesting for industrial processing, where the frequency, phase, and amplitude should be stabilized in an economical fashion. Fujii *et al.* [8] demonstrated a magnetron that was externally locked and provided a gain of 26 dB. Furthermore, Shinohara and Matsumoto [9] and Tahir *et al.* [10] successfully achieved a narrowband amplifier that delivered a power gain of 30–40 dB using injection-locked magnetrons. The gains of SSPAs and klystrons may easily achieve in excess of 50 dB [11]. Thus, it is attractive to study injection-locked magnetrons to obtain a lower injection signal power, wider locking bandwidth, and purer output spectrum. Greater amplitude and phase stability are expected to improve the reproducibility in industrial applications, and reduce the interference with Wi-Fi or Bluetooth equipment via frequency hopping. Furthermore, the power of a single magnetron does not currently generally meet industrial requirements, but

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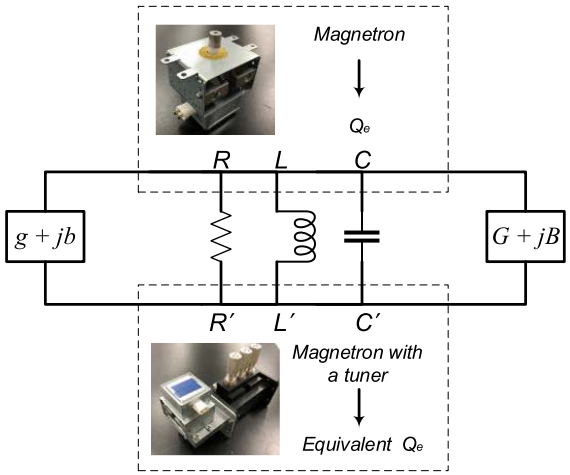


Fig. 1. Equivalent parallel *RLC* resonance circuit of a magnetron.

combining multiple magnetrons to achieve a high output power with high efficiency requires phase control and a wide locking bandwidth [12], [13].

This paper presents a high power gain *S*-band magnetron amplifier based on a load–pull characterization and an investigation of the locking bandwidth, stability, and spectral purity of the locked magnetron amplifier. Additionally, a microwave power combination system was investigated based on the proposed dual magnetron amplifier.

II. Theory and Analysis

The magnetron oscillator can be described by a single-mode equivalent parallel *RLC* resonance circuit [14], as shown in Fig. 1. The magnetron resonance cavity is equivalent to the lumped *RLC* circuit; the characteristics of electronic discharge and electronic interactions are represented by the conductance g and susceptance b, and the external load is described by the load conductance G and susceptance B.

According to Slater [14], the instantaneous amplitude and frequency of the free-running magnetron are given as

$$V_{\mathrm{MW}} = \frac{V_{\mathrm{dc}}}{1 + \frac{C\omega_0 R}{Q_L}} \tag{1}$$

$$\omega_f = \omega_0 + \frac{b_0}{2C} - \frac{B}{2Q_e}\omega_0 - \frac{\omega_0 \tan\alpha}{2Q_L} \tag{2}$$

where V_{dc} and V_{MW} are the dc voltage across the anode–cathode gap and the magnetron output microwave voltage, respectively; Q_L and Q_e are the external and loaded quality factors of the magnetron; b_0 is a constant; α is the frequency pushing parameter; C and R are the capacitor and resistor in the equivalent lumped *RLC* circuit, respectively; $\omega_0 = \sqrt{LC}$ is the resonance frequency and ω_f is the instantaneous angular frequency of the magnetron.

Obviously, (1) and (2) indicate that the amplitude and frequency of a free-running magnetron are proportional to V_{dc}, and the magnetron amplitude and frequency fluctuate with the comparatively slow fluctuations of V_{dc}. In order to reduce such fluctuations, an in-house-designed dc analog stabilized power supply maintained a voltage ripple of less than 1.5‰ was built in the system. Additionally, a free-running magnetron presents a narrower spectral width via reducing the high voltage ripple of power supply [15]. Thus, the high dc voltage with low ripple is a basic factor to achieve the amplitude stability, frequency stability, and injection locking [16].

When the external weak reference signal is injected into the magnetron, the amplitude, frequency, and phase of the injection locking magnetron are derived with Chen's mode [17]

$$\frac{d\theta}{dt} = \omega_0 \frac{\rho}{2Q_e}\sin\theta + \omega_f - \omega_{\mathrm{inj}} \tag{3}$$

$$V_{\mathrm{MWinj}} = V_{\mathrm{MW}} \frac{1}{1 + \frac{\rho\cos\theta}{Q_e/Q_0 + G/2}} \tag{4}$$

where $\rho = (P_{\mathrm{inj}}/P_{\mathrm{out}})^{1/2}$ is the injection ratio, $\theta = (\omega_f - \omega_{\mathrm{inj}})t + \varphi$, φ is the phase difference between the reference signal and the output of the free-running magnetron, and t is the time.

Consequently, the injection locking condition can be analytically solved from (3) by setting $d\varphi/dt = 0$; we thus obtain the well-known Adler's condition [18]

$$\frac{\rho\omega_0}{2Q_e} \geq |\omega_f - \omega_{\mathrm{inj}}|. \tag{5}$$

The block diagram of a 1-kW continuous wave (CW) magnetron amplifier system based on WR430 waveguides is shown in Fig. 2. The reference plane (a dotted line in the right side of a three-stub tuner shown in Fig. 2) was defined to analyze the proposed magnetron system. The magnetron with three-stub tuner in series was recognized as an equivalent external excited resonant cavity, which generates high-power signals and acts as a load to the injection signal. Therefore, the proposed system was equivalent to the basic injection model by introducing the equivalent Q_e. With such conditions, the behaviors of the proposed magnetron system can be evaluated by the above-mentioned basic injection locking theory, as shown in Fig. 1.

The Q_e factor of the whole excited resonant cavity was calculated with S-parameter from the cold-test model of the magnetron resonator [19]. The relative position of three-stub tuner's pins was shown in Fig. 2, and the depths of three-stub tuner's pins were tuned from 0 to 5, 10, and 15 mm with respect to the inner waveguide wall, respectively. The computed cold-test results of the equivalent Q_e factor with load–pull characterization were shown in Table I.

Apparently, Table I depicted that the decreased equivalent Q_e factor and increased coupling with the magnetron can be obtained with the load–pull characterization simultaneously. Equation (5) illustrates that the injection power can be decreased, or the locking range can be extended substantially by decreasing the Q_e factor of the injection-locked magnetron. Thus, the injection locking is easier and more stable.

Chosen original state ($Q_e = 110$) and load–pull lower Q_e state ($Q_e = 59$) for illustrating the conclusion mentioned earlier, the injection-locked bandwidth is measured. As shown in Fig. 3, the bandwidth was measured at various injection ratios for comparison. Considering the load–pull characterization, when the Q_e factor was tuned to 59, the magnetron supplied nearly twice locking bandwidth of an original injection-locked magnetron. It was the first time that the

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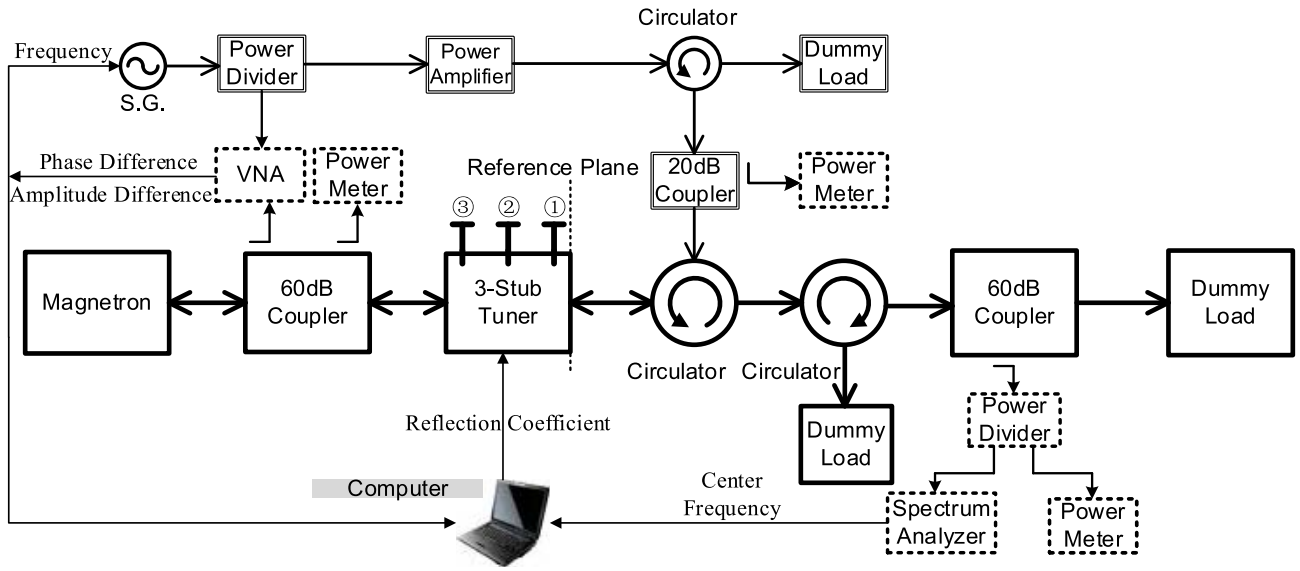


Fig. 2. Diagram of the 1-kW CW magnetron amplifier system (dashed framework part: measurement setup; upper part: injection signal route; Middle highlight path: high-power output route; and connections to computer: control signal route).

TABLE I
COMPUTED COLD TEST RESULTS WITH DIFFERENT TUNING STUB DEPTHS

Tuning Stub	Depth (mm)	Frequency (GHz)	R_L(dB)	Q_e
No Tuner	0	2.45510	-2.95	110
①	5	2.449092	-3.82	110
①	10	2.449092	-3.52	59
①	15	2.710383	-4.67	47
②	5	2.449092	-3.82	110
②	10	2.449092	-4.58	137
②	15	2.446054	-8.01	350
③	5	2.449092	-3.82	110
③	10	2.449092	-4.34	101
③	15	2.755606	-5.77	47

Frequency is the resonant frequency of equivalent cavity, and R_L is the return loss of equivalent cavity at the resonant frequency.

$$Q_e = \frac{f_0}{\Delta f}\left(1+\frac{1-S_0}{1+S_0}\right);\quad R_L = -20\lg\left[(1+S_0)/(1-S_0)\right]$$

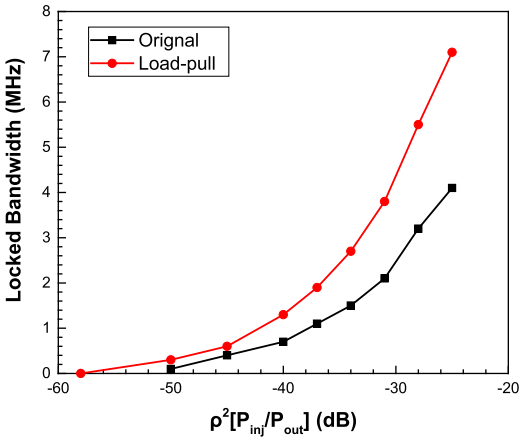


Fig. 3. Injection-Locked bandwidth with various injection ratios at different states.

magnetron had been locked at a gain of 58.0 dB, while the original injection gain was only 50.0 dB.

As a result, to construct a magnetron amplifier system, a low equivalent Q_e-factor region can achieve relatively high gain, wide bandwidth, and increase the coupling to the magnetron, which enhances the injection locking as well [18]. It is noted that a balance between the injection ratio and output stability is necessary in the injection-locked magnetron system, because the higher the injection ratio, the more stable the output amplitude and phase are achieved with a small-signal injection [20].

When the injection-locked magnetron satisfied Adler's condition, the injection-locked magnetron spectral density of the phase noise can be represented as the following Pengvanich's method [21]:

$$|\delta\tilde{\theta}(\omega)|^2 = \frac{|\delta\tilde{\omega}_f(\omega)|^2}{\omega^2 + \left(\frac{\rho}{2Q_e}\right)^2 - \left(\frac{\omega_f-\omega_0}{\omega_0}\right)^2} \tag{6}$$

where $\delta\theta(\omega)$ is the spectral density of the phase noise and $\delta\omega_f(\omega)$ is the fluctuation at the free-running frequency.

Specifically, due to the increased coupling, which is provided by decreasing R_L shown in Table I, the magnetron with load–pull characterization receives more injection power, and the injection ratio increases. Thus, we could derive from (1) and (4) that a decreased equivalent Q_e factor and an enhanced coupling to the magnetron increase the output voltage of the injection-locked magnetron. Then, (6) illustrates that under the same condition, the output spectrum and phase noise are improved as well. Meanwhile, (6) shows that the phase jitter can be estimated by $\delta\omega_f(\omega)/\omega$, and the 0.016 rad jitter would achieved in the proposed system. Our previous patent on the injection-locked magnetron microwave generator with recycle of spurious energy [22] also supports the aforementioned behavior of the injection-locked magnetron.

On the other hand, (5) indicates that the locking of a magnetron is easily achieved when $(\omega_f - \omega_{\text{inj}})$ is negligibly small, so controlling the instantaneous angular frequency of

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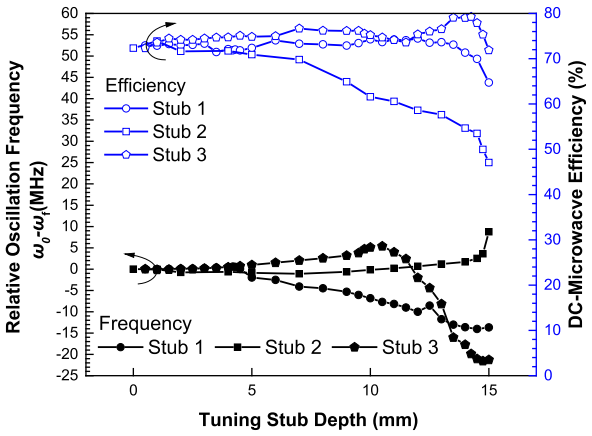


Fig. 4. Frequency and efficiency of free-running magnetron with various tuning stub depths.

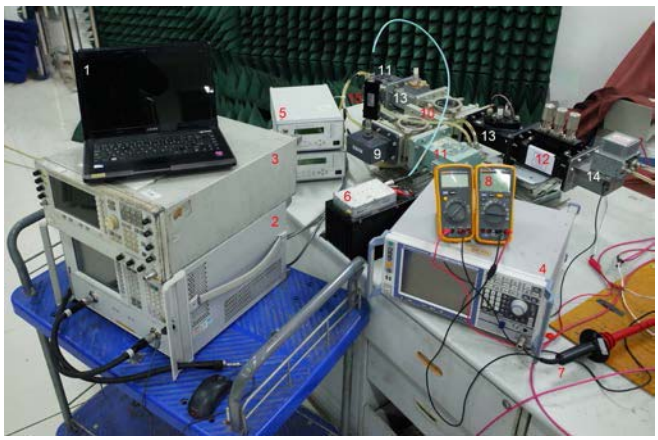


Fig. 5. Hardware setup for the 1-kW CW magnetron amplifier system (1: computer; 2: vector network analyzer; 3: signal generator; 4: signal analyzer; 5: power meter; 6: power amplifier with isolators; 7: high-voltage probe; 8: Fluke multimeter; 9: waveguide-coaxial converter; 10: circulators; 11: water-cooled dummy loads; 12: three-stub tuner; 13: 60-dB coupler; 14: magnetron and external excitation cavity; and 15: 20-dB coupler).

the magnetron is another way to enhance the bandwidth of the magnetron. The previous frequency pushing method has already been demonstrated experimentally [10]; in that method, the output power of the magnetron changes obviously when the magnetron is external injection locked at different frequencies. Fortunately, (2) indicates that another frequency pulling ($B\omega_0/Q_e$) element is included in an instantaneous free-running frequency. It also states that the instantaneous free-running frequency changes when either B or Q_e changes. Meanwhile, it is greatly contributed to the proposed method that the instantaneous frequency of the magnetron is independently controlled without affecting the output power.

To explain the above-mentioned relationship, the relative position of three-stub tuner's pins was shown in Fig. 2, and the depths of three-stub tuner's pins were tuned from 0 to 5, 10, and 15 mm with respect to the inner waveguide wall again, respectively. Results of the center frequency (the center frequency is normalized by $\omega_{f0} = 2.4510$ GHz) and the efficiency of the free-running magnetron with the load–pull characterization were measured as shown in Fig. 4. Such results were consistent with the previous cold-test equivalent Q_e factors (Table I): when Q_e decreased, the free-running frequency and efficiency increased and vice versa. The free-running frequency shifted 33.0 MHz and the highest efficiency reached 80% (original efficiency: 70.0%). Meanwhile, it was certain that the free-running frequency varied in the low-equivalent Q_e factor regions, so that both the bandwidth and efficiency were improved by the load–pull method.

The phase difference between the injection signal and the output signal of the magnetron is a constant in the steady state, as shown in (7) [14], [17], [20]. Equation (7) points out that the phase θ between the magnetron's output and the external reference signal varies when the magnetron's natural frequency ω_f drifts or injection frequency changes at the locking state. Hence, if the injection locking magnetron works for a long time, both frequency and load–pull characterization are controlled to stabilize its amplitude and phase. It is possible to control the magnetron output phase with load–pull characterization, since the free-running frequency ω_f depends on frequency pulling element, which is shown in Fig. 4

$$\theta = \arcsin\left[\frac{(\omega_f - \omega_{\text{inj}}) \cdot 2Q_e}{\rho\omega_0}\right]. \tag{7}$$

Thus, the load–pull method is one of the best ways to achieve the desired locking process. The frequency, output power, and output phase can all be controlled independently using the proposed method. There is no doubt that if the load characterization is pulled to the low Q_e-factor regions and optimized, the injection locking magnetron can operate as an excellent linear narrowband amplifier with $G = P_{\text{out}}/P_{\text{inj}}$ defined as the power gain and $2\Delta\omega$ defined as the bandwidth. Simultaneously, the behavior of the magnetron amplifier (injection locking bandwidth, injection ratio, efficiency, and phase noise) is also expected to be improved.

III. Experimental System

We designed and built a 1-kW CW magnetron amplifier system, the main hardware components of which are shown in Fig. 5 without their detailed measurement connections and without the electronically steerable modules of the modified three-stub tuner.

The household appliance magnetron used for the experiments was the Panasonic 2M210-M1 with an in-house-designed dc analog stabilized power supply without a particular bias voltage control. This power supply maintained a voltage ripple of less than 1.5‰ when the magnetron was the load. A signal generator (E8267C, Agilent) amplified with a 50-dB power amplifier (ZHL-30W-262, Mini-Circuits) provided the low-power-level injection signal to the magnetron. The circulators with the dummy loads transmitted the injected signal to the magnetron and acted as isolators to protect both the magnetron and the amplifier. The output microwave power from the magnetron was mainly absorbed by water cooled dummy loads (TMWGWLC10, Euler, Chengdu, China). The peak frequency of the injected reference signal and the free-running magnetron output were sampled to calculate the

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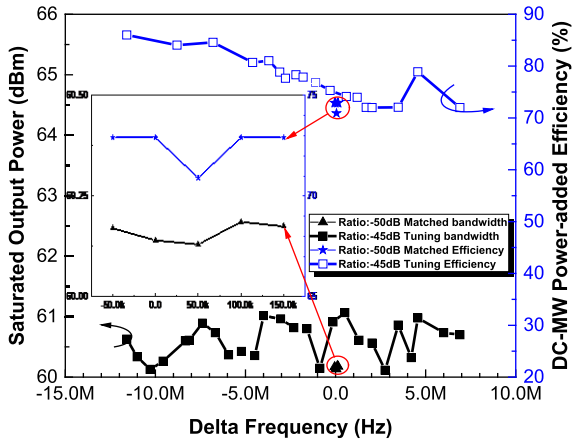


Fig. 6. Output and locked bandwidth of the magnetron amplifier with matched or tuning loads, respectively.

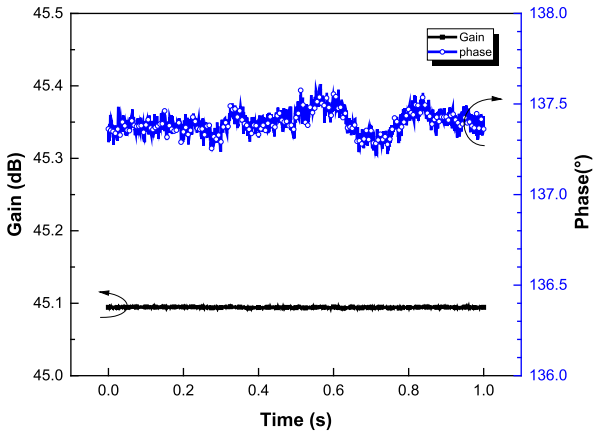


Fig. 7. Amplitude and phase jitter of the magnetron amplifier with a 45-dB power gain.

differences between these frequencies; the frequency difference was decreased by tuning the magnetron reflection coefficient (i.e., via a load–pull characterization of the magnetron) with the help of a modified three-stub tuner (HD-22WST3, HD Microwave Co., Xi'an, China). Finally, with careful tuning, the required impedance of the magnetron was obtained, and the magnetron was successfully locked at the reference frequency; the required output phase was also achieved. Three directional couplers (LOOP22DC60D20C 1010N, Euler) were employed to sample the microwave signals for the power (power meter, AV2433, the 41st Institute of CETC), amplitude, and phase measurements (vector network analyzer, E8362B, Agilent), as well as for the spectrum and phase noise measurements (signal analyzer, FSV40, Rohde & Schwarz).

The center frequency of the free-running magnetron oscillated at 2.4510 GHz with an output power of 60.0 dBm (anode voltage of −3806 V and anode current of 0.360 A). Once the free-running oscillation started, the cathode current was turned off to reduce the frequency modulation noise, which stabilized the oscillation to yield a narrowband spectrum [23].

IV. Experimental Results

According to the definition of the power added efficiency in a traditional SSPA, we defined the efficiency of the 1-kW CW magnetron amplifier system here as follows:

$$\text{Eff}(\%) = \frac{P_L - P_{\text{inj}}}{P_{\text{dc}}} \tag{8}$$

where P_L is the power delivered to the water cooler dummy load, P_{inj} is the power of the injected signal, and P_{dc} is the dc power supplied to the magnetron.

The gain bandwidth and efficiency characteristics of the optimized magnetron amplifier were evaluated and are presented in Fig. 6. A bandwidth of 0.20 MHz was achieved for the direct injection-locked magnetron at an output to the injection power ratio ($P_{\text{out}}/P_{\text{inj}}$) of 50 dB, which agreed with predictions using Alder's condition [18]; furthermore, the bandwidth could be significantly extended to 18.5 MHz when the magnetron amplifier employed the load–pull characterization. Thus, it could deliver a power gain of 45 dB for a limited operating magnetron reflection power of less than 50.4 dBm (output voltage standing wave ratio = 2). The magnetron amplifier achieved an output power above 60.0 dBm with an output power flatness of less than 1.0 dB; additionally, as the input injection power contributes increasingly to the output power at lower power gains, the output power increased as the power gain $P_{\text{out}}/P_{\text{inj}}$ increased. An excellent dc to microwave power-added efficiency was also measured for the injection-locked magnetron amplifier (it was above 72.0% across the entire locked bandwidth). The fluctuation of the efficiency was mainly caused by a small amount of drift of the anode current via the three-stub tuner tuning.

Furthermore, if no restrictions were placed on both the magnetron reflection power and the efficiency, then a magnetron amplifier tuning locked bandwidth of more than 30.0 MHz could be achieved with a 45-dB gain; additionally, the amplitude of the output signal could be set by controlling the anode current.

A vector network analyzer was used to monitor the amplitude and phase jitter between the magnetron amplifier output and the injected reference signal. Fig. 7 shows the measured amplitude and phase jitter of the locked magnetron output at around 1.0 s. The peak-to-peak amplitude was nearly constant (less than 0.01 dB) and the peak-to-peak phase shift of the magnetron output was reduced from ±180° (free-running mode) to 0.5° (injection locking mode) peak to peak, and it was well coincide with the previous estimation. We also demonstrated that the phase jitters of the injection-locked magnetron in the current system can be attributed to fluctuations of the dc power supply, because the period of the phase jitters and the rectified mains ripple remained consistent. Most importantly, once the magnetron was locked with the 45-dB power gain, a maximum magnetron output phase control range of almost 180° could be achieved via three-stub tuner tuning, while the amplitude and frequency could be independently stabilized. This is the first time that the magnetron output phase control has been successfully demonstrated based on the load–pull characterization at a power gain of 45 dB.

Fig. 8(a) shows the spectrum of the free-running and locked magnetron's output characteristics. From Fig. 8(a), it can be seen that the random frequency chirp is visible in the characteristics during free-running operation and that the

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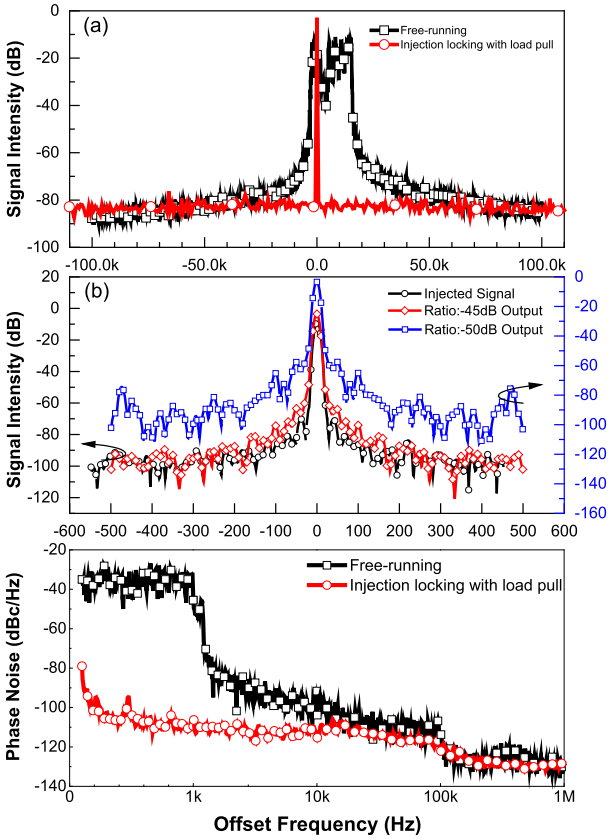


Fig. 8. Spectrum and phase noise of the magnetron (of 10 Hz).

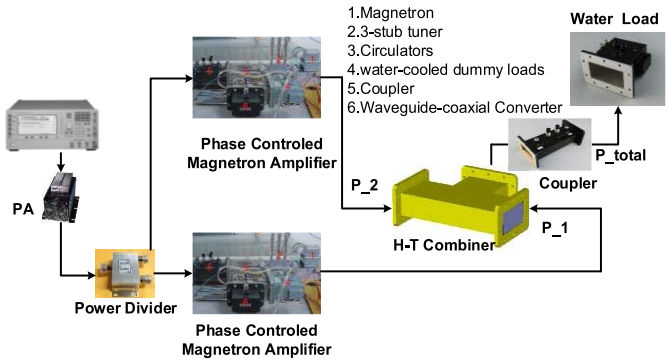


Fig. 9. Diagram of the dual magnetron amplifiers used for microwave power combining.

TABLE II
COMPARISON OF PREVIOUSLY DESIGNED INJECTION-LOCKED MAGNETRONS

Source	Power Gain (dB)	Bandwidth	Phase Noise (dBc/Hz)	Stability (peak-to-peak)
[5]	30	—	-59@59 Hz	0.84% / 0.73 °
[8]	26	within 5 MHz	-70@40 kHz	—
[9][23]	30	>6 MHz	-85@10 kHz	1.0% / 1.0°
[10]	40	2Mb/s for PSK	-40@100 kHz	—
[25]	30	15 MHz	—	—
This work	45	18.5 MHz	-106.3@1 kHz	0.25% / 0.5°

locked magnetron amplifier showed no frequency variation and a relatively sharp frequency spectrum. This low noise performance of the free-running oscillators and the appropriate reflection are the key factors to easily achieve injection locking [16]. Furthermore, Fig. 8(b) shows a close-up view of the magnetron spectrum at different P_{out}/P_{inj} ratios compared with the injected reference signal. When the P_{out}/P_{inj} power ratio was 50 dB, the central part of the spectrum of the oscillation was a little wider than that of the reference signal; it should be noted that two pairs of microphonic signals appeared, both the −64.2-dB microphonic signal (spurious suppression equal to −69 dBc) at ±100 Hz and the −74.1-dB microphonic signal at ±500 Hz were caused by rectified mains ripples. By decreasing the P_{out}/P_{inj} ratio to 45 dB, the microphonic signal became smaller and the locked magnetron output spectrum could be fully synchronized with the reference signal and the spectral purity of the locked magnetron could be significantly improved. Finally, Fig. 8(c) shows the measured average phase noise of the locked magnetron with a power gain of 45 dB and the free-running magnetron. The output phase noise improved visibly and was measured to be −106.3 dBc/Hz at 1 kHz, which is a significant improvement compared with values reported in the literature [5], [8]–[10]. The main reason in this paper choosing a magnetron amplifier that could deliver a power gain of 45 dB for this system was for its excellent noise performance at such a P_{out}/P_{inj} ratio.

Owing to the physical limitations of the single magnetron's power capacity, the power output does not satisfy the requirements for high-power industrial applications. Based on the aforementioned magnetron amplifier, coherent power combining of dual 1-kW magnetron amplifiers has been readily achieved using an H-plane tee power combiner. The diagram of the microwave power combining system is shown in Fig. 9. The signal is provided by an Agilent signal generator (E8267C) and amplified with a power amplifier; then, the amplified signal is split into two equiphase, equiamplitude parts for injection into the dual magnetrons. Two 60-dB directional couplers were employed to sample the input microwave signals of the H-plane tee power combiner for the power measurements, while another 60-dB directional coupler was employed to measure the microwave power combining output.

Both magnetrons were locked at 2.4510 GHz and P_{out}/P_{inj} was about 45 dB. The powers of the dual magnetrons delivering to the H-plane tee power combiner were 1027.0 and 999.0 W. The average output microwave power reached 1839.6 W. The successfully obtained average power combining efficiency was higher than 90.8% and the phase noise degradation could be ignored in the total output signal; furthermore, there was no any additional phase shifter in our power combining system. Thus, the injection-locked magnetron amplifier shows potential for use as an inexpensive but high-gain, high power amplifier for industrial applications [24].

Additionally, this paper can be evaluated better, a comparison with various already reported designs is presented in Table II. To the best of the authors' knowledge, this is one of the highest power gains (45 dB) and phase noise performances (−106.3 at 1 kHz) ever reported for an *S*-band injection-locked magnetron with a power output of over 1 kW.

V. CONCLUSION

A high power gain magnetron amplifier with output phase control and a relevant phase-shifterless power combining

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system have been developed and measured. The equivalent resonant cavity method was first introduced to facilitate the proposed magnetron amplifier system as a basic injection model, and the system behavior was qualitatively explained using an equivalent parallel *RLC* resonance circuit model. Then, with the help of the load–pull characterization, it is the first time to demonstrate a magnetron that has been locked at a gain of 58 dB. A wider locked bandwidth (18.5 MHz) and better output phase control range were obtained, while the power-added efficiency was enhanced by 72%. This is the first time a successful magnetron output phase control based on load–pull at a power gain of 45 dB has been demonstrated. An excellent output amplitude stability of 0.01 dB (peak to peak), a good phase stability of 0.5° (peak to peak), and low phase noise of −106.3 dBc/Hz at 1-kHz offset were achieved. It is also demonstrated that the proposed dual 1-kW magnetron amplifier employed for coherent power combining is highly efficient without introduced any extra phase shifter equipment. The results show that an injection-locked magnetron has potential to be applied as an inexpensive, high-gain power amplifier for industrial applications.

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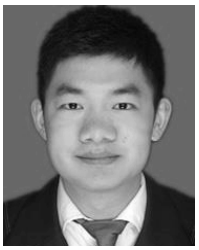
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